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CELOVAN
Vancomycin
500 mg
Powder for Solution for Injection/Infusion

PRODUCT DESCRIPTION:

The product Vancomycin powder for solution for injection /infusion 500 mg is a white to off white lyophilized plug or powder. The reconstituted solution is a clear, colourless solution.

COMPOSITION:

Each vial contains: Vancomycin Hydrochloride (lyophilized) equivalent to Vancomycin 500 mg.

PHARMACOLOGY:

Vancomycin is a glycopeptide antibiotic with bactericidal action against a variety of Gram-positive bacteria. It is not active in-vitro against gram- negative bacilli, mycobacteria or fungi. It exerts its action by inhibiting cell wall biosynthesis of susceptible bacteria. There is also evidence that Vancomycin alters the permeability of the cell membrane and selectively inhibits ribonucleic acid synthesis. There is no cross-resistance between Vancomycin and other antibiotics.

PHARMACOKINETICS:

Vancomycin is poorly absorbed after oral administration. It is given intravenously for therapy of systemic infections as intramuscular injection is painful.

The mean elimination half-life of Vancomycin from plasma is 4 to 6 hours in subjects with normal renal function. In the first 24 hours, about 75% of an administered dose of Vancomycin is excreted in urine by glomerular filtration. Mean plasma clearance is about 0.058 L/kg/hr, and mean renal clearance is about 0.048 L/kg/hr. Renal dysfunction slows excretion of Vancomycin. In anephric patients, the average half-life of elimination is 7.5 days. The distribution coefficient is from 0.3 to 0.43 L/kg.

There is no apparent metabolism of the drug. About 60% of an intraperitoneal dose of Vancomycin administered during peritoneal dialysis is absorbed systemically in six hours. Serum concentrations of about 10 mcg/mL are achieved by intraperitoneal injection of 30 mg/kg of Vancomycin. Although Vancomycin is not effectively removed by either hemodialysis or peritoneal dialysis, there have been reports of Vancomycin clearance with haemoperfusion and haemofiltration. Total systemic and renal clearance of Vancomycin may be reduced in the elderly.

Vancomycin is approximately 55% serum protein bound as measured by ultrafiltration at Vancomycin serum concentrations of 10 to 100 mcg/mL. After I.V. administration of Vancomycin, inhibitory concentrations are present in pleural, pericardial, ascitic, and synovial fluids; in urine, in peritoneal dialysis fluid and in atrial appendage tissue. Vancomycin does not readily diffuse across normal meninges into the spinal fluid; but, when the meninges are inflamed, penetration into the spinal fluid occurs.

INDICATIONS:

Vancomycin Hydrochloride is indicated in the following:

- Serious or severe infections caused by susceptible strains of methicillin-resistant (ß-lactam- resistant) staphylococci.
- Staphylococcal endocarditis.
- Infections caused by staphylococci including septicaemia, bone infections, lower respiratory tract infections, skin and skin structure infections.
- Endocarditis caused by Streptococcus viridans, S.bovis or enterococci E. faecalis (Vancomycin used alone or in combination with an aminoglycoside)
- Diphtheroid endocarditis.
- Pseudomembranous colitis caused by C. difficile
- Staphylococcal enterocolitis.

DOSAGE:

Vancomycin is recommended to be administered by intermittent IV infusion for the treatment of systemic infections. It is very irritating to the tissue and must not be given IM. To minimize adverse effects, concentrations of no more than 5 mg/mL and rates of no more than 10 mg/min are recommended in adults. In selected patients in need of fluid restriction, a concentration up to 10 mg/mL may be used; use of higher concentrations may increase the risk of infusion-related adverse effects. Infusion-related adverse effects may occur, however, at any rate or concentration.

Reconstitution & administration:

Dissolve 500 mg in 10 ml of Sterile Water for Injection to give a solution of 50 mg/mL. Reconstituted solutions containing 500 mg of Vancomycin must be further diluted with at least 100 ml of diluent. The desired dose should be administered by intermittent IV infusion over a period of at least 60 minutes.

Compatibility with IV Fluids: 5% Dextrose Injection; 0.9% Sodium Chloride Injection.

Patients with Normal Renal Function:

Adults: The usual IV dose is 2 g daily, divided either as 500 mg every 6 hours or 1 g every 12 hours. Factors such as age and obesity in patients may need modification of the usual daily IV dose.

Children: The usual IV dosage is 10mg/kg/dose given every 6 hours.

Infants and Neonates: The recommended initial dosage of 15 mg/kg, followed by 10 mg/kg every 12 hours for neonates in the first week of life and every 8 hours for infants from 1 week to 1 month of age. Close monitoring of serum concentrations may be warranted in these patients.

Patients with impaired renal function:

In patients with impaired renal function, including premature infants and elderly patients, greater dosage reductions than expected may be necessary because of decreased renal function. Measuremant of Vancomycin serum concentrations can be helpful in optimizing therapy, especially seriously ill patients with changing renal function.

For functional anephric patients, an initial IV dose of 15 mg/kg of body weight should be given to achieve prompt therapeutic serum concentrations. Subsequent dosage must be based mainly on renal function and serum concentrations of the drug.

SIDE EFFECTS:

Infusion-related: Anaphylactoid reactions during or soon after rapid infusion. Reactions such as hypotension, wheezing, dyspnea, urticaria, pruritus, flushing of the upper body (“red neck”) or pain and muscle spasm of the chest and back, usually resolve within 20 mins but may persist for several hours. Such side-effects are infrequent if Vancomycin is given by a slow infusion over 60 mins or at a rate of 10 mg/min or less.

Front Side

ARTWORK DETAIL LABEL

Product	**CELOVAN – Vancomycin 500 mg**				
Buyer/Country	Mylan / Malaysia	Component	Pack Insert		
Dimension	180 x 200 mm			Pack	--
New Item Code	**1034295**	Old Item Code	1031238		
Colour Shades	Black			No. of Colours	1

Change Control No.	PR # 3151594
Design/Style	Front & Back Printing. To be supplied in folded size of 55 x 35 mm. Brand name facing front after final folding.
Substrate	40/45 GSM Paper.
Special Instructions	Printing clarity to be clear & sharp.
Autocartonator Requirements	NA
Caution to the printer: Before processing, please ensure that the ARTWORK received for printing is exactly in line with APPROVED ARTWORK provided to you. In case of any FONTS/DESIGN are Mis-matching with the APPROVED ARTWORK, please inform PDC for further action. **DO NOT MAKE ANY CHANGE TO THE ARTWORK WITHOUT WRITTEN INSTRUCTIONS FROM PDC.**	

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Nephrotoxicity: Renal failure, principally manifested by increased serum creatinine or BUN concentrations, especially in patients given large doses of Vancomycin, has rarely been reported.

Ototoxicity: Hearing loss has been reported mostly in patients who had kidney dysfunction or a preexisting hearing loss or were receiving concomitant treatment with an ototoxic drug. Vertigo, dizziness and tinnitus have rarely been reported.

Hematopoietic: Reversible neutropenia, thrombocytopenia, reversible agranulocytosis.

Miscellaneous: Phlebitis, anaphylaxis, drug fever, nausea, chills, eosinophilia, rashes, Steven-Johnson syndrome and vasculitis.

PRECAUTIONS:

To reduce development of drug-resistant bacteria, the drug should be used only for the treatment or prevention of infections proven or strongly suspected to be caused by susceptible bacteria.

Prolonged use of Vancomycin may result in the overgrowth of non-susceptible organisms. Careful observation of the patient is essential. If superinfection occurs during therapy, appropriate measures should be taken. In rare instances, there have been reports of pseudomembranous colitis due to C. difficile developing in patients who received IV Vancomycin.

In order to minimize the risk of nephrotoxicity when treating patients with underlying renal dysfunction or patients receiving concomitant therapy with an aminoglycoside, serial monitoring of renal function should be performed and particular care should be taken in following appropriate dosing schedules. Serial tests of auditory function may be helpful in order to minimize the risk of ototoxicity.

Reversible neutropenia has been reported in patients receiving Vancomycin. Patients who will undergo prolonged therapy with Vancomycin or those who are receiving concomitant drugs which may cause neutropenia should have periodic monitoring of the leukocyte count.

Thrombophlebitis may occur, the frequency and severity of which can be minimized by administering the drug slowly as a dilute solution (2.5 to 5 g/L) and by rotating the sites of infusion. There have been reports that the frequency of infusion-related adverse reactions (including hypotension flushing, erythema, urticaria and pruritus) increases with the concomitant administration of anesthetic agents. These reactions may be minimized by the administration of Vancomycin as a 60 minutes infusion prior to the anesthetic induction.

The safety and efficacy of Vancomycin administration by the intraperitoneal and intrathecal (intralumbar or intraventricular) routes have not been assessed.

USE IN PREGNANCY AND LACTATION:

Pregnancy: Study of Vancomycin in pregnant women is limited. It is not known whether the drug can cause foetal harm when administered to pregnant women. Vancomycin should be used during pregnancy only when clearly needed.

Nursing Mothers: Vancomycin is excreted in human milk. Caution should be exercised when Vancomycin is administered to a nursing woman. Because of the potential for adverse events, a decision should be made whether to discontinue nursing or to discontinue the drug, taking into account the importance of the drug to the mother.

CONTRAINDICATIONS:

Vancomycin is contraindicated in patients with known hypersensitivity to this antibiotic.

DRUG INTERACTIONS:

Concomitant administration of Vancomycin and anesthetic agents has been associated with erythema and histamine-like flushing, and anaphylactic reactions.

Because of the possibility of additive toxicities, the concurrent, sequential systemic or topical use of other ototoxic and/or nephrotoxic drugs (e.g., aminoglycosides, amphotericin B, bacitracin, cisplatin, colistin, polymyxin B) and Vancomycin requires careful serial monitoring of renal and auditory function. These drugs should be used with caution in patients receiving Vancomycin therapy.

OVERDOSE AND TREATMENT:

Supportive care is advised, with maintenance of glomerular filtration. Vancomycin is poorly removed by dialysis; however, haemofiltration and haemoperfusion using polysulfone resin reportedly has been of limited value.

AVAILABILITY:

500 mg vials: Box of 1 vial / Box of 10 vials

STORAGE:

Store below 30°C. Protect from light. Keep Medicine Out of Reach of Children.

Vancomycin Powder for Solution for Injection/Infusion 500 mg is stable for 48 hours when stored at room temperature (below 25°C) and 2-8°C when diluted individually with water for injection, with 0.9% Sodium Chloride and with 5% Dextrose.

For further information, please consult your physician or pharmacist.

Mylan

Product Registration Holder in Malaysia:
MYLAN HEALTHCARE SDN. BHD.
15-03 & 15-04, Level 15, Imazium,
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Manufactured by:
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Bangalore - 560 105, India.

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